

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Name	Kanmani L
Register Number	192511414

MODEL PRACTICAL EXAMINATION – SET 3

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Employee, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Net Pay, Gross Pay, etc.
2. Create a simple cloud software application for Students Attendance System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Students, Course, Present, Absent, Course Attendance, OD Request, Grade, Attendance Percentage, etc.
3. Demonstrate virtualization by installing a Type-2 Hypervisor on your device; create and configure a VM image with a Host Operating System (either Windows/Linux).
4. Create a Virtual Hard Disk and allocate the storage using VMware Workstation.

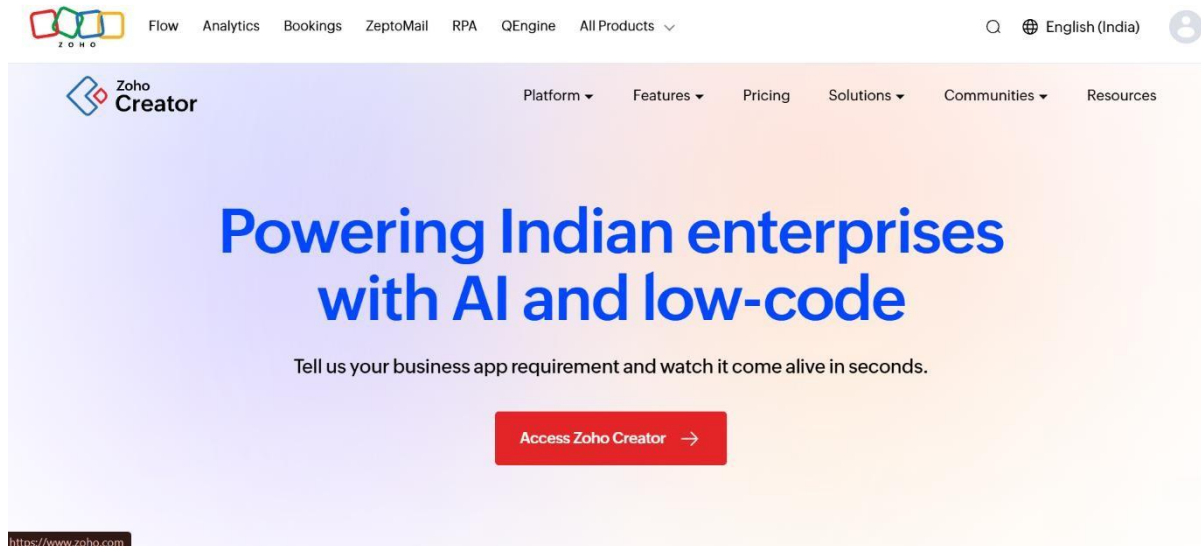
1. Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Employee, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Net Pay, Gross Pay, etc.

AIM:

To create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

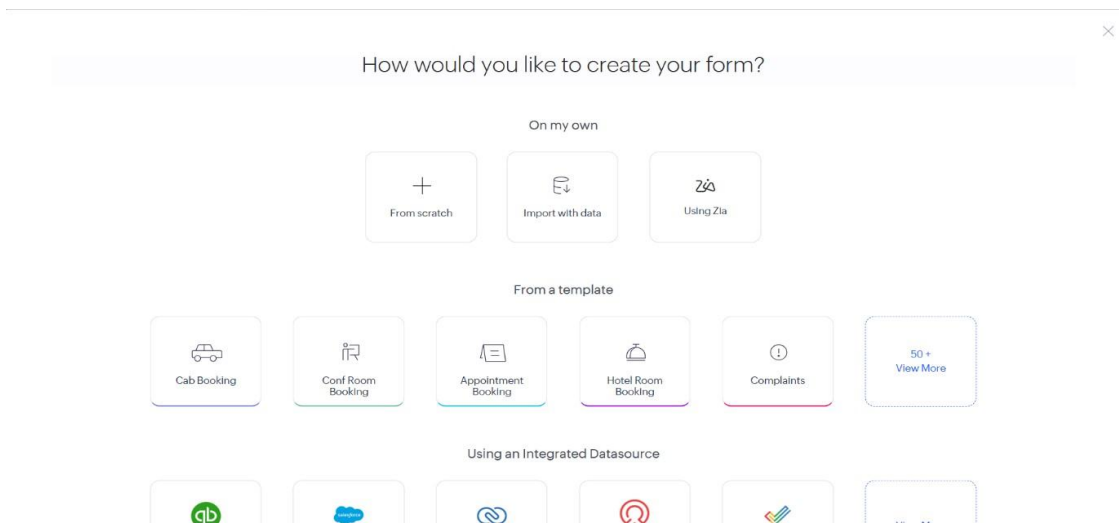
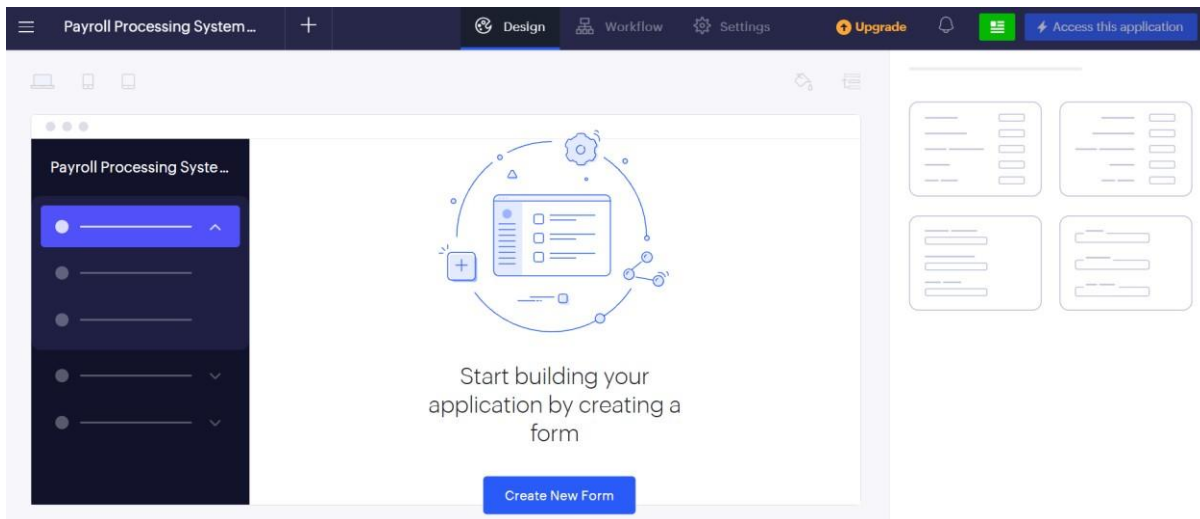
STEP 1: Log into zoho.com.



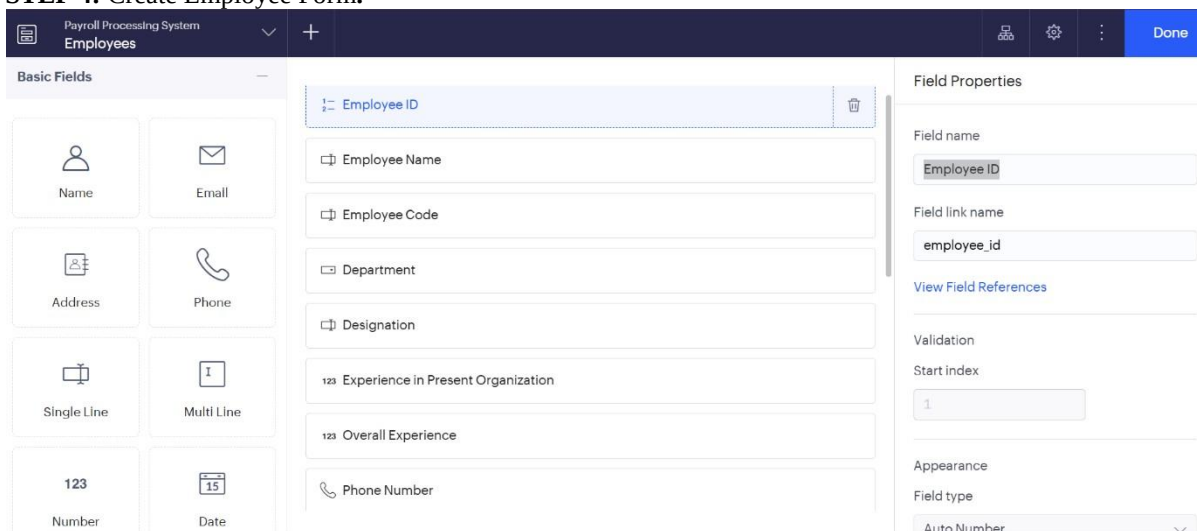
STEP 2: Create a new application from scratch in Zoho Creator.

The image shows the "Create from scratch" form in Zoho Creator. At the top, there is a logo consisting of four colored circles (blue, green, yellow, red) arranged in a square. Below the logo is the text "Create from scratch". The form has a light gray background. It contains a text input field labeled "Application Name" with a red asterisk indicating it is required. The input field contains the text "Payroll Processing System". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" next to it. At the bottom of the form is a large blue button labeled "Create Application".

STEP 3: Create a new application and create forms from scratch.



STEP 4: Create Employee Form.



STEP 5: Create Payroll records Form.

Payroll Processing System

Payroll Records

+

🖨️

⚙️

⋮

Done

Basic Fields

👤

Name

✉️

Email

🏠

Address

☎️

Phone

📄

Single Line

📄

Multi Line

123

Number

📅

Date

🔍

Payroll ID

🗑️

🔍

Employee

📅

Salary Month

💰

Basic Salary

💰

Bonus

💰

Deductions

💰

Gross Salary

💰

Net Salary

Field Properties

Field name

Payroll ID

Field link name

payroll_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

Get started faster!

Help

STEP 6: Create Payroll Records form

Payroll Processing System

Payment Records

+

🖨️

⚙️

⋮

Done

Basic Fields

👤

Name

✉️

Email

🏠

Address

☎️

Phone

📄

Single Line

📄

Multi Line

123

Number

📅

Date

🔍

Employee

🗑️

🔍

Payroll

📅

Payment Date

📄

Payment Method

📄

Payment Status

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employees

Employee ID

Display type

Multi Select

STEP 7: Create Leave records form.

Payroll Processing System

Leave Records

+

🖨️

⚙️

⋮

Done

Basic Fields

👤

Name

✉️

Email

🏠

Address

☎️

Phone

📄

Single Line

📄

Multi Line

123

Number

📅

Date

🔍

Employee

🗑️

📄

Leave Type

📅

From Date

📅

To Date

📄

Leave Status

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employees

Employee ID

Display type

Multi Select

STEP 8: Add sample data to all the created forms.

Payroll Processing System
Trial expires in 14 days
Upgrade
Edit this application
Help

Payroll Processing System
Dashboard
Payment Records
Leave Records
+ Leave Records
Leave Records Rep...
Board
Payroll Records
Employees

Leave Records

Employee: 4 x

Leave Type: Sick Leave x v

From Date: 28-Jul-2026

To Date: 29-Jul-2026

Leave Status: Approved x v

Submit
Reset

Payroll Processing System
Trial expires in 14 days
Upgrade
Edit this application
Help

Payroll Processing System
Dashboard
Payment Records
+ Payment Records
Payment Records ...
Board
Leave Records
Payroll Records
Employees

Payment Records

Employee: 3 x

Payroll: 4 x

Payment Date: 28-Jul-2026

Payment Method: Cash x v

Payment Status:
☐ Paid
☒ Pending
☐ Other

Submit
Reset

STEP 11: Create reports to display the data entered in the respective forms.

Payroll Processing System
Trial expires in 14 days
Upgrade
Edit this application
Help

Payroll Processing System
Dashboard
Payment Records
+ Payment Records
Payment Records ...
Board
Leave Records
Payroll Records
Employees

Payment Records Report

	Employee	Payroll	Payment Date	Payment Method	Payment Status
	3	4			
	1	1	28-Jul-2026	Bank Transfer	Paid
	2	2			
		3			
		4			
	2	2	28-Jul-2026	UP	Paid
		3			
		4			
...	1	1	28-Jul-2026	Cash	Paid
	2	3			
	3				
	3	4	28-Jul-2026	Cash	Pending

Showing 6 of 6

Payroll Processing System

Trial expires in 14 days

Upgrade

Edit this application

Help

Payroll Processing System

Dashboard

Payment Records

Leave Records

Leave Records

Leave Records Rep...

Board

Payroll Records

Employees

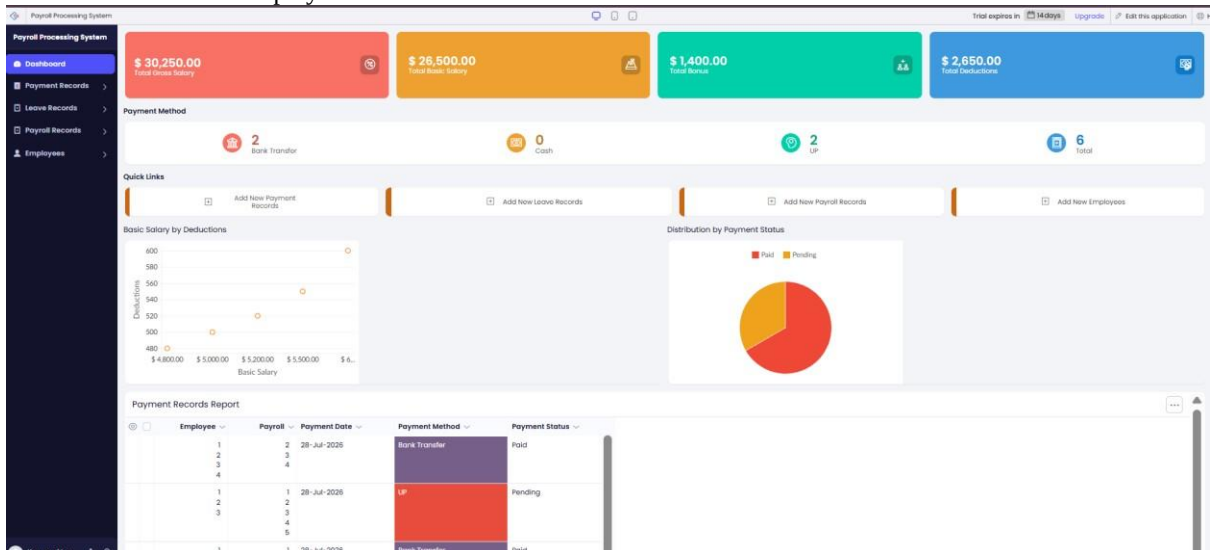
Kanmani Log...

Leave Records Report

Employee	Leave Type	From Date	To Date	Leave Status
4	Sick Leave	28-Jul-2026	29-Jul-2026	Approved
1	Casual Leave	25-Jul-2026	28-Jul-2026	Rejected
2				
3				
1	Emergency Leave	20-Jul-2026	22-Jul-2026	Approved
2				
3				
1	Sick Leave	10-Jul-2026	12-Jul-2026	Pending
1	Casual Leave	01-Jul-2026	05-Jul-2026	Approved
2				

Showing 5 of 5

STEP 12: Create and display the Dashboard



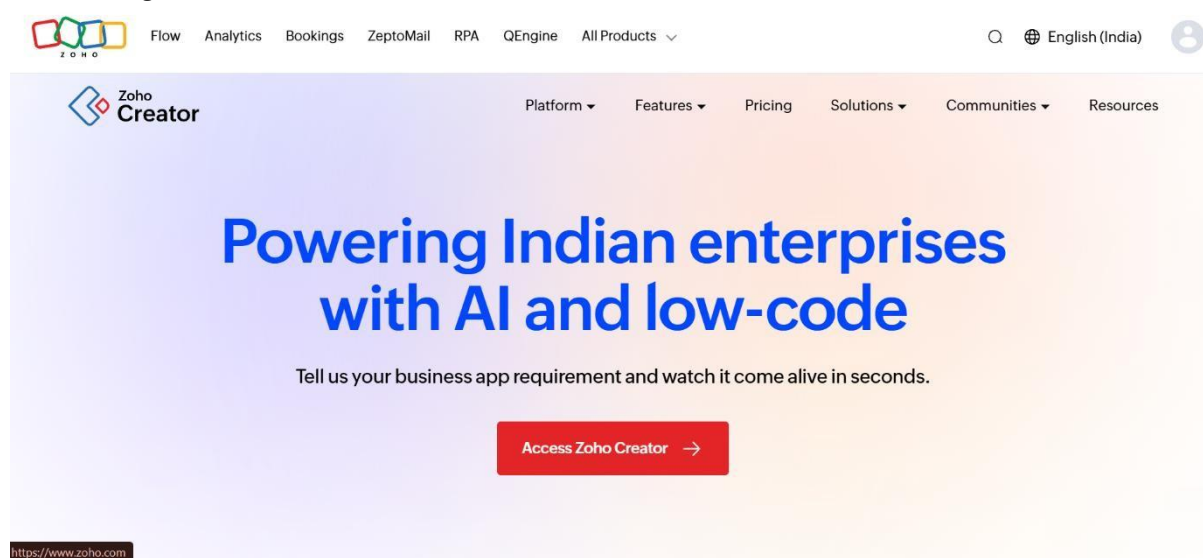
2. Create a simple cloud software application for Students Attendance System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Students, Course, Present, Absent, Course Attendance, OD Request, Grade, Attendance Percentage, etc.

AIM:

To create a simple cloud software application for Students Attendance System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

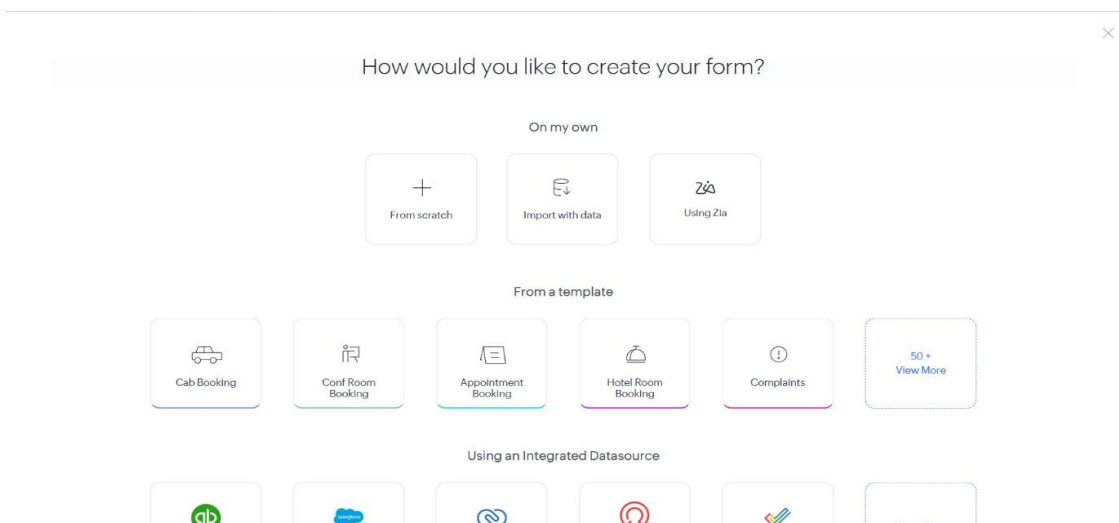
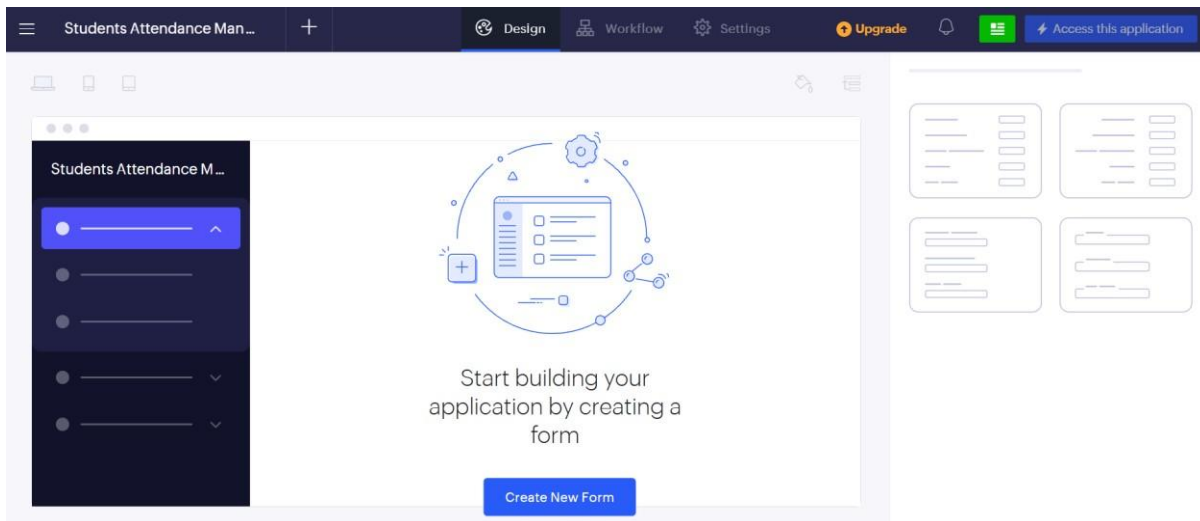
STEP 1: Log into zoho.com.



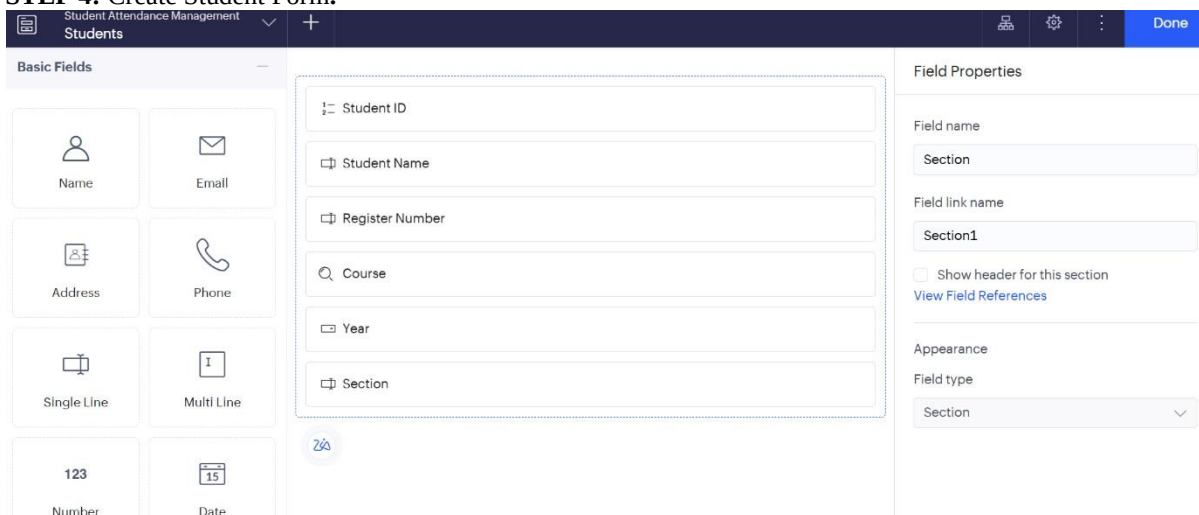
STEP 2: Create a new application from scratch in Zoho Creator.

A screenshot of the "Create from scratch" form in Zoho Creator. At the top, there are four icons: a blue circle, a white square, a white square with a plus sign, and a white square with a plus sign. Below the icons is the text "Create from scratch". The form contains a label "Application Name" with a red asterisk, followed by a text input field containing "Students Attendance Management System". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" in blue. At the bottom of the form is a large blue button labeled "Create Application".

STEP 3: Create a new application and create forms from scratch.



STEP 4: Create Student Form.



STEP 5: Create Course Form.

Student Attendance Management

Courses

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Course ID

Course Code

Course Name

Faculty Name

Semester

2/3

Field Properties

Field name

Course ID

Field link name

course_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

STEP 6: Create Attendance Summaries Form.

Student Attendance Management

Attendance Summaries

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Student

Course

123 Total Classes

123 Classes Attended

.00 Attendance Percentage

Grade

2/3

Field Properties

Field name

Student

Field link name

student

View Field References

Display Fields

Student Attendance Management → Students

Student ID

Display type

Multi Select

STEP 7: Create Attendance Entries

Student Attendance Management

Attendance Entries

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Student

Course

Date

Status

.00 Hours Attended

2/3

Field Properties

Field name

Student

Field link name

student

View Field References

Display Fields

Student Attendance Management → Students

Student ID

Display type

Multi Select

STEP 8: Create OD Requests Form.

Student Attendance Management
OD Requests

Basic Fields

Name Email

Address Phone

Single Line Multi Line

123 Date

Student

Course

OD Date

Reason

Status

Field Properties

Field name: Student

Field link name: student

View Field References

Display Fields

Student Attendance Management → Students

Student ID

Display type: Multi Select

STEP 9: Create Feedback Form.

Student Attendance Management
Feedbacks

Basic Fields

Name Email

Address Phone

Single Line Multi Line

123 Date

Student

Comments

Field Properties

Field name: Student

Field link name: student

View Field References

Display Fields

Student Attendance Management → Students

Student ID

Display type: Multi Select

STEP 10: Add sample data to all the created forms.

Student Attendance Management

Trial expires in 14 days Upgrade Edit this application

Students

Student Name: Shubnum

Register Number: 192509037

Course: 2

Year: II

Section: c

Submit Reset

Dashboard

Attendance Entries

Attendance Sum...

Students

+ Students

Students Report

Board

Courses

OD Requests

Feedbacks

Kanmani Log...

STEP 11: Create reports to display the data entered in the respective forms.

Student Attendance Management

Trial expires in 14 days

Upgrade

Edit this application

Student Attendance Mana...

Dashboard

Attendance Entries

Attendance Sum...

Students

Students

Students Report

Board

Courses

OD Requests

Feedbacks

Kanmani Log...

Students Report

Student ID

Student Name

Register Number

Course

Year

Section

6	Shubnum	192509037	2	II	c
5	Eva Garcia	STU2026005	1	III	B
4	David Lee	STU2026004	2	II	C
3	Carol Martinez	STU2026003	1	III	A
2	Bob Williams	STU2026002	1	IV	B
1	Alice Johnson	STU2026001	2	I	A

Showing 6 of 6

STEP 12: Create and display the Dashboard with student details, course attendance, present and absent records, OD requests, attendance percentage, grades, and overall performance.

Student Attendance Management

Trial expires in 14 days

Upgrade

Edit this application

Student Attendance Mana...

Dashboard

Attendance Entries

Attendance Sum...

Students

Students

Students Report

Board

Courses

OD Requests

Feedbacks

Kanmani Log...

8.50

Average of Hours Attended

\$25.00

Total Courses

91.13

Average of Attendance Percentage

5

Count of Attendance Entries

Grade

1

Excellent

0

Poor

2

Average

2

Good

Quick Links

Add New Attendance Entries

Add New Attendance Summaries

Add New Students

Add New Courses

Grade by Attendance Percentage

Good

Excellent

Average

0

50

100

150

Distribution by Status

Absent

Present

Attendance Entries Report

Student	Course	Date	Status	Hours Attended
1	1	23-Jul-2026	Present	9.1
2	2	24-Jul-2026	Present	9.1
3	2	25-Jul-2026	Present	7.1
4	1	26-Jul-2026	Absent	9.1
1	1	27-Jul-2026	Absent	9.1

3. Demonstrate virtualization by installing a Type-2 Hypervisor on your device; create and configure a VM image with a Host Operating System (either Windows/Linux).

Aim:

To demonstrate Type-2 virtualization by installing VMware Workstation and creating a configured Ubuntu virtual machine.

Procedure:

Step 1: Install and open VMware Workstation on the host operating system.

Step 2: Select Create a New Virtual Machine and choose the Typical configuration option.

Step 3: Select the Ubuntu ISO image and provide a suitable name for the virtual machine.

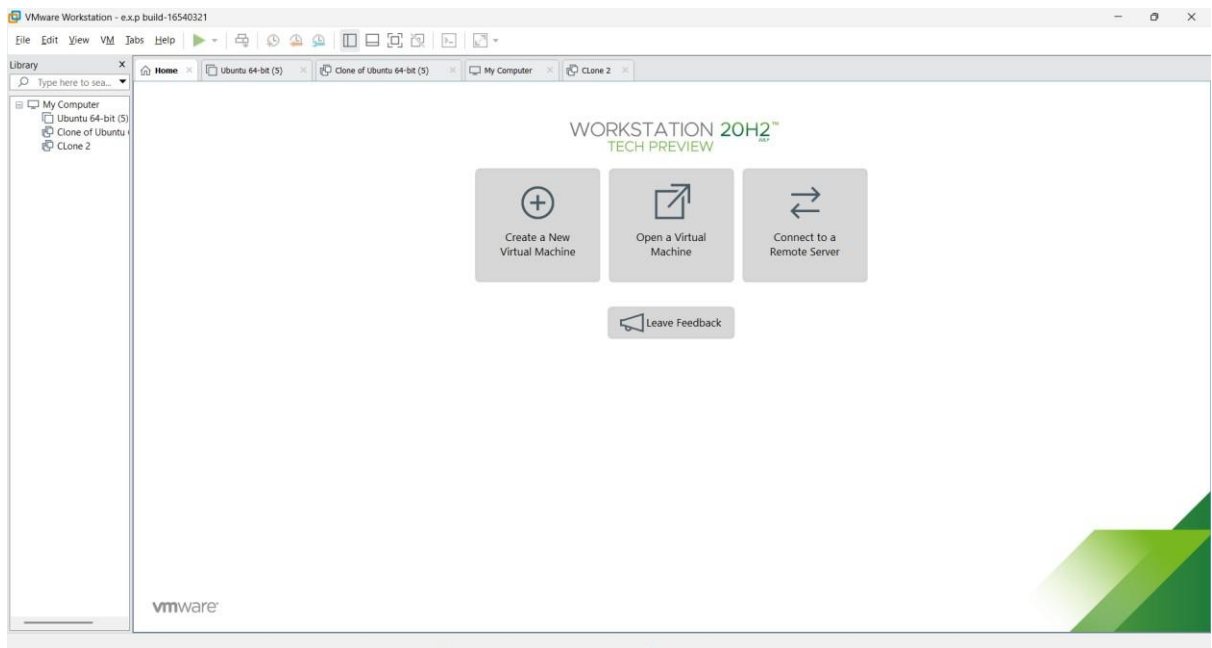
Step 4: Open Customize Hardware and configure 1 CPU, 2 GB RAM, and 15 GB disk storage.

Step 5: Select NAT for the network adapter and complete the virtual machine configuration.

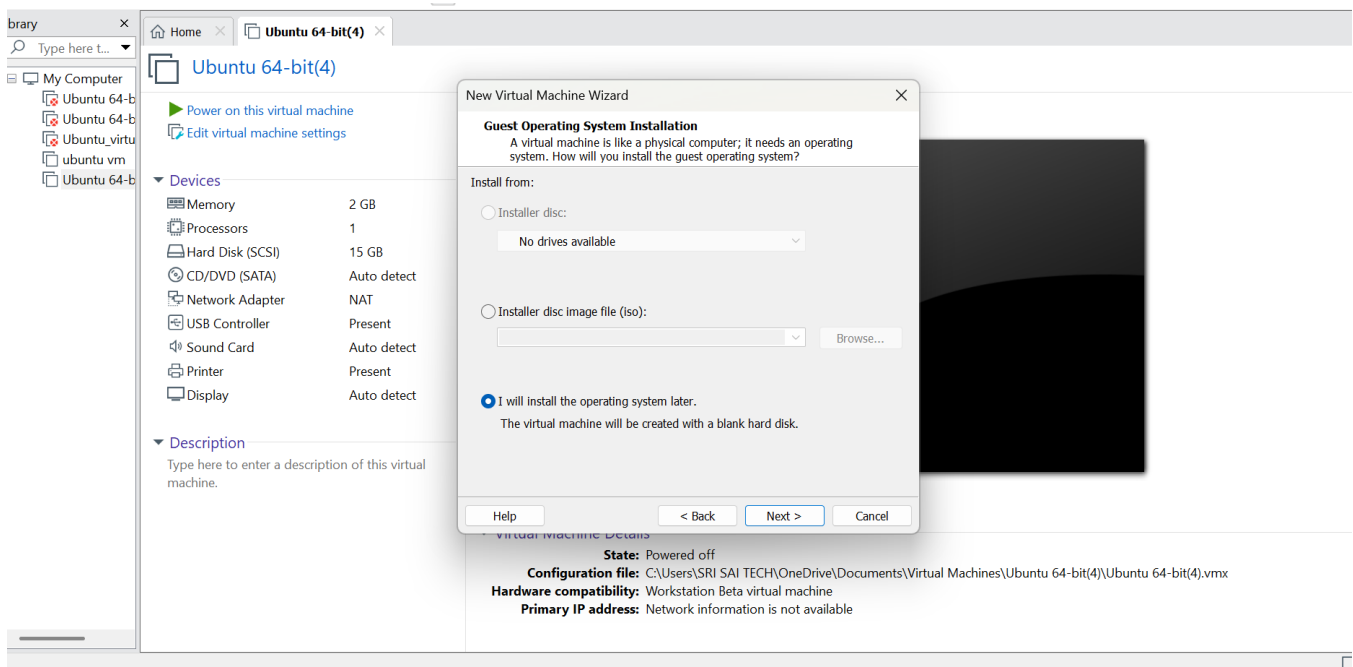
Step 6: Start the VM, install Ubuntu, and verify that the guest operating system runs successfully inside VMware Workstation.

IMPLEMENTATION:

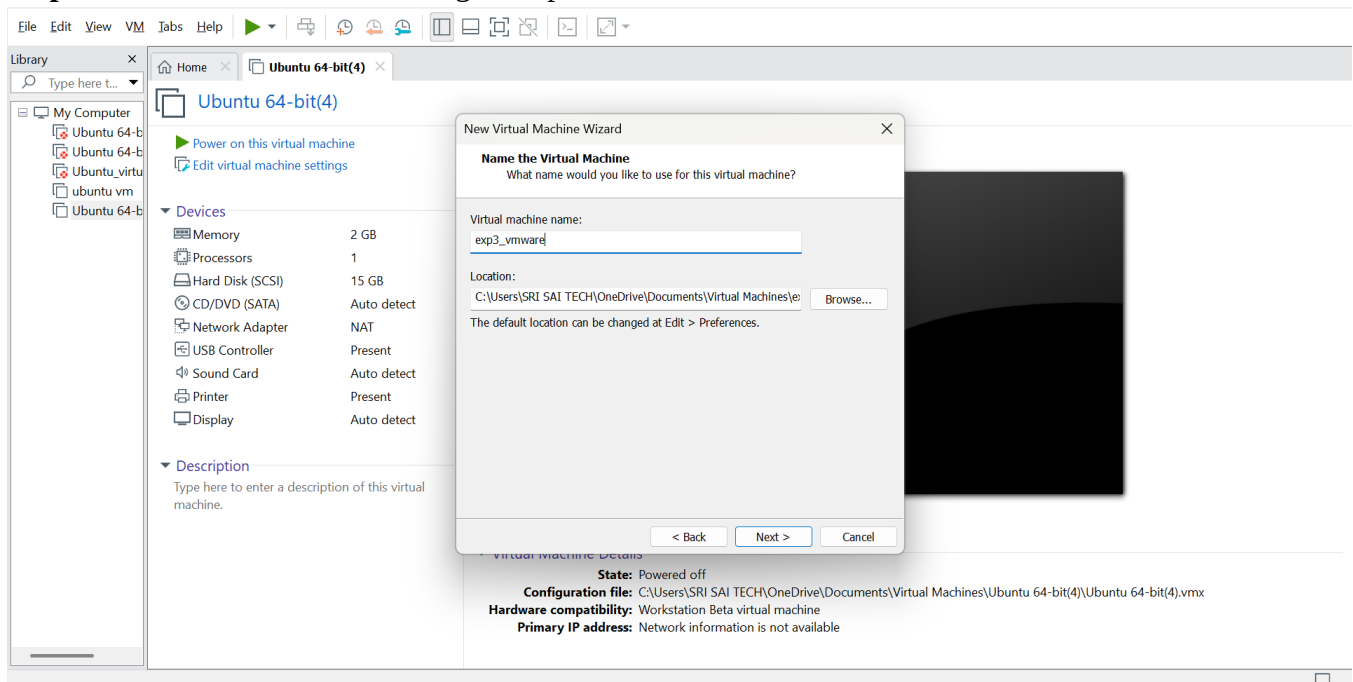
Step 1: Install and open VMware Workstation on the host operating system.



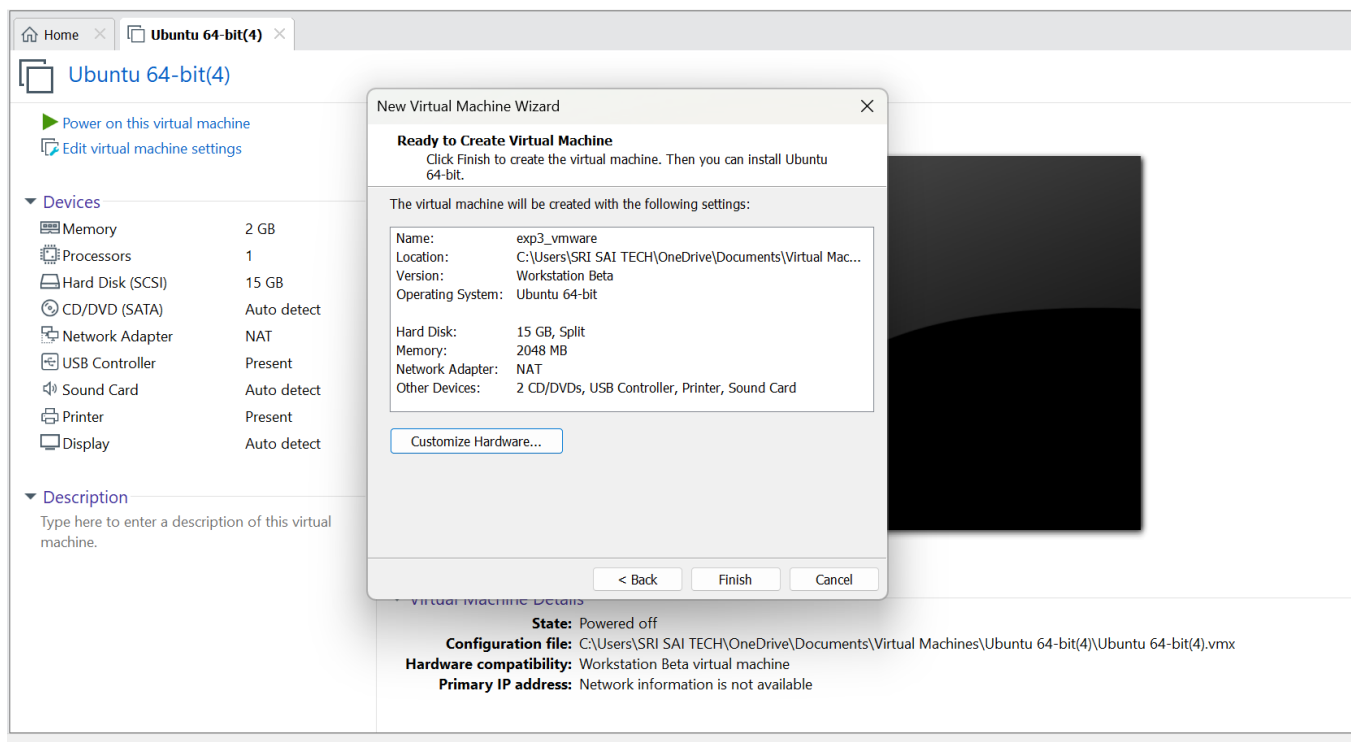
Step 2: Select Create a New Virtual Machine and choose the Typical configuration option.



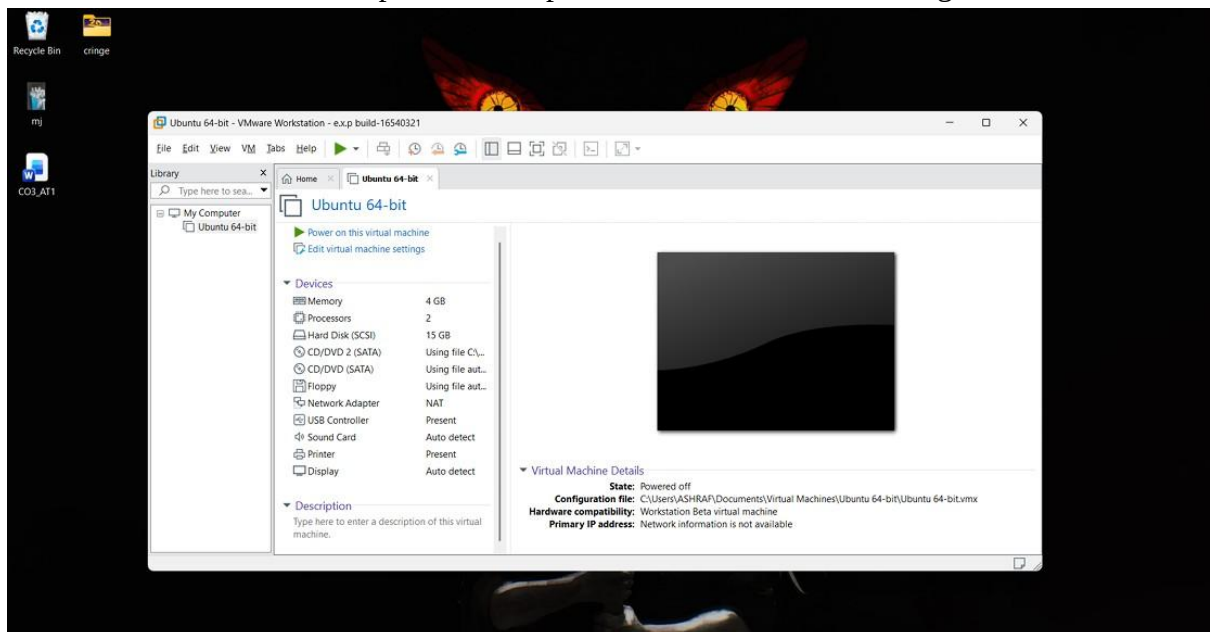
Step 3: Select the Ubuntu ISO image and provide a suitable name for the virtual machine.



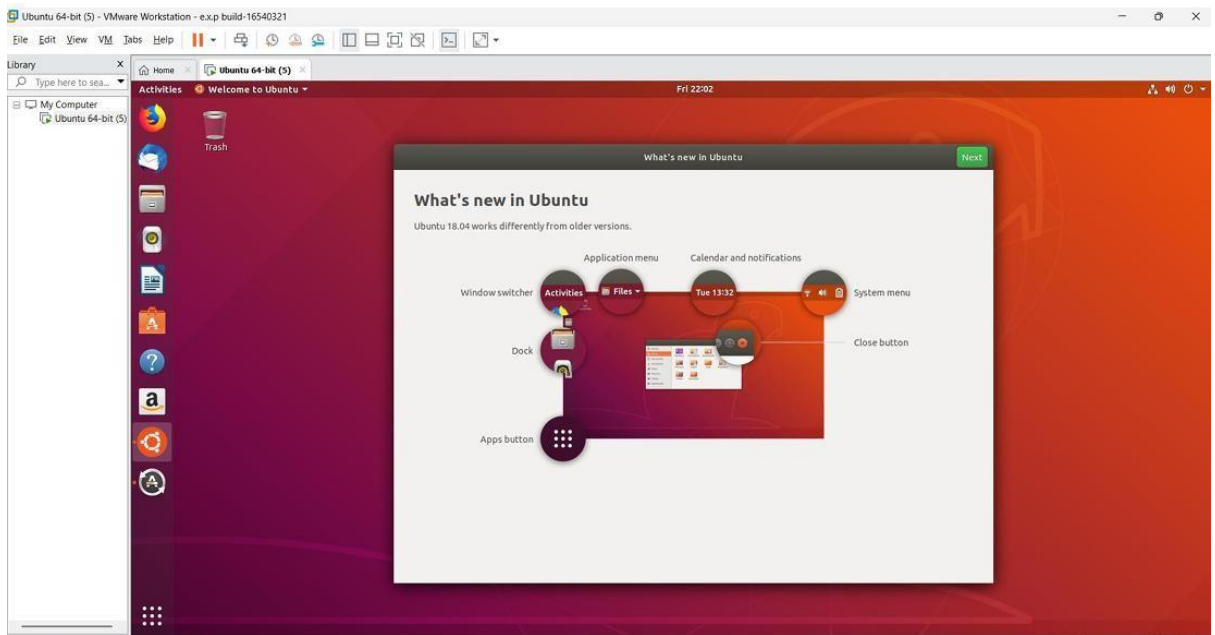
Step 4: Open Customize Hardware and configure 1 CPU, 2 GB RAM, and 15 GB disk storage.



Step 5: Select NAT for the network adapter and complete the virtual machine configuration.



Step 6: Start the VM, install Ubuntu, and verify that the guest operating system runs successfully inside VMware Workstation.



4. Create a Virtual Hard Disk and allocate the storage using VMware Workstation.

Aim:

To create a virtual hard disk and allocate storage to it using VirtualBox.

Procedure:

Step 1: Open VMware Workstation and select the existing Ubuntu virtual machine.

Step 2: Click Edit virtual machine settings.

Step 3: Select Hard Disk and click Add to create a new virtual hard disk.

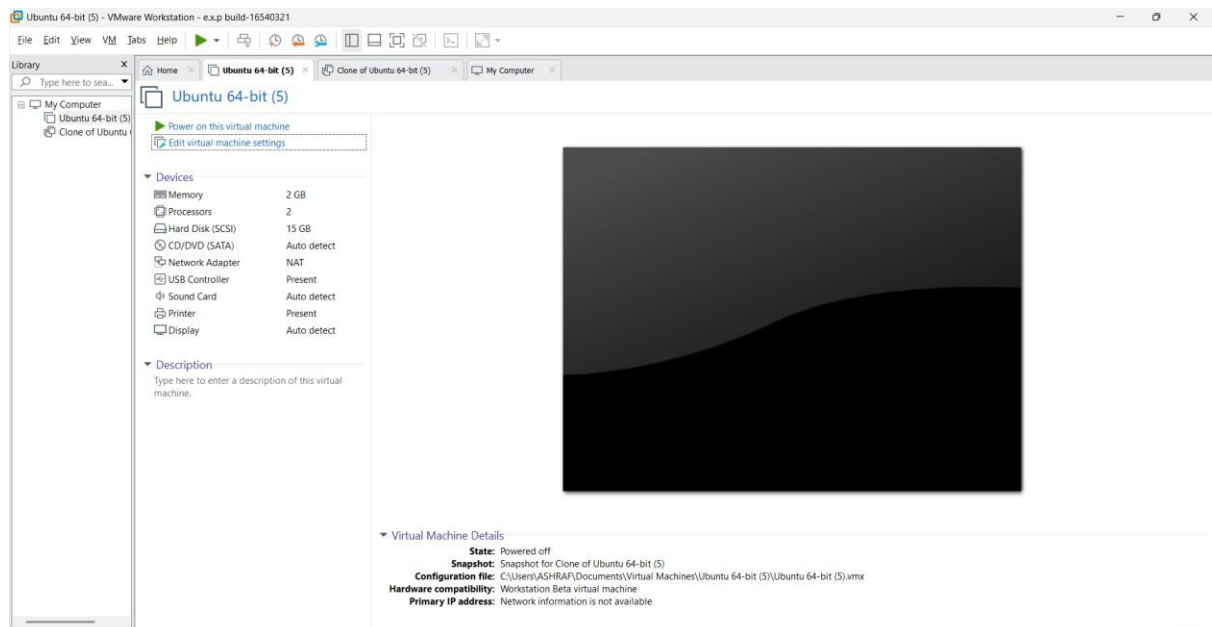
Step 4: Select Hard Disk → SCSI as the virtual disk type.

Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.

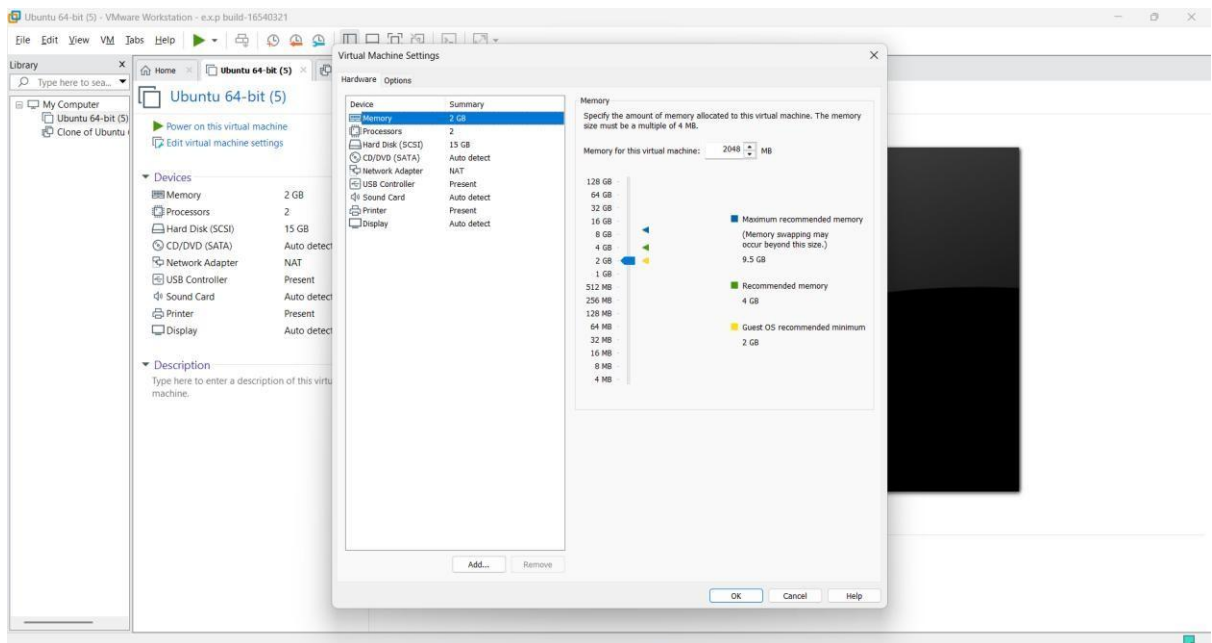
Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.

IMPLEMENTATION:

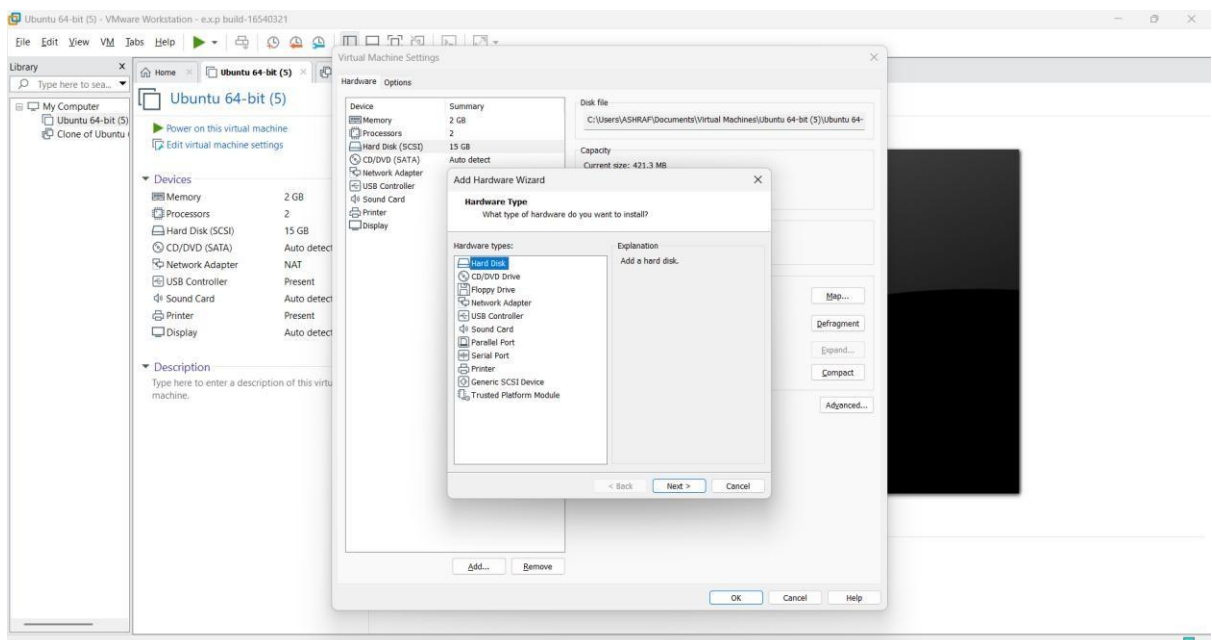
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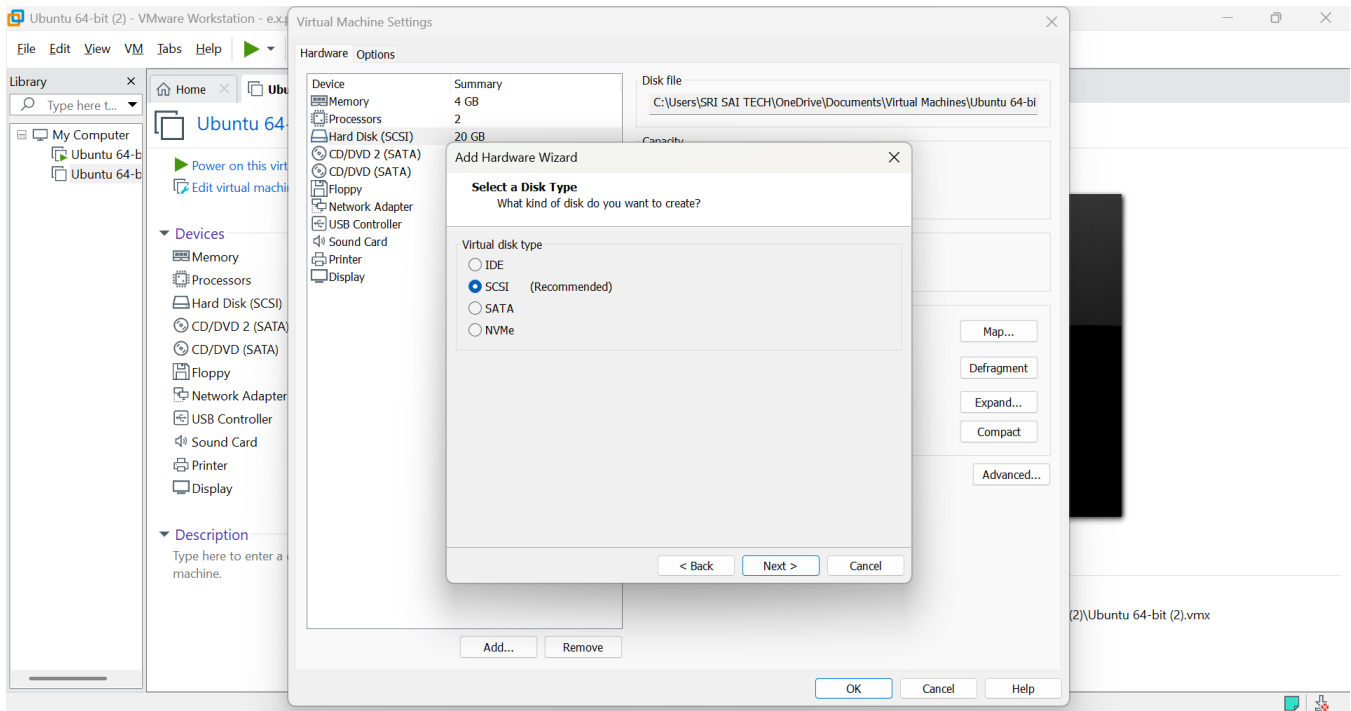
Step 2: Click Edit virtual machine settings.



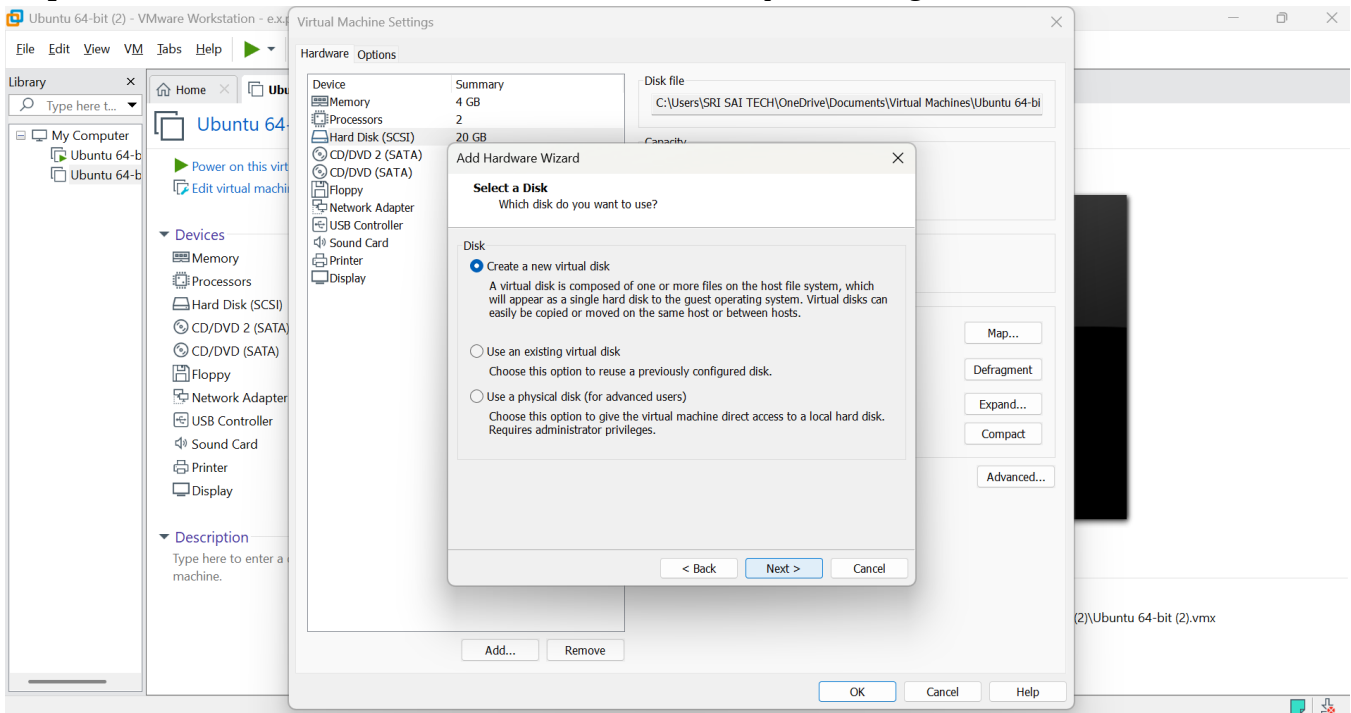
Step 3: Select Hard Disk and click Add to create a new virtual hard disk.

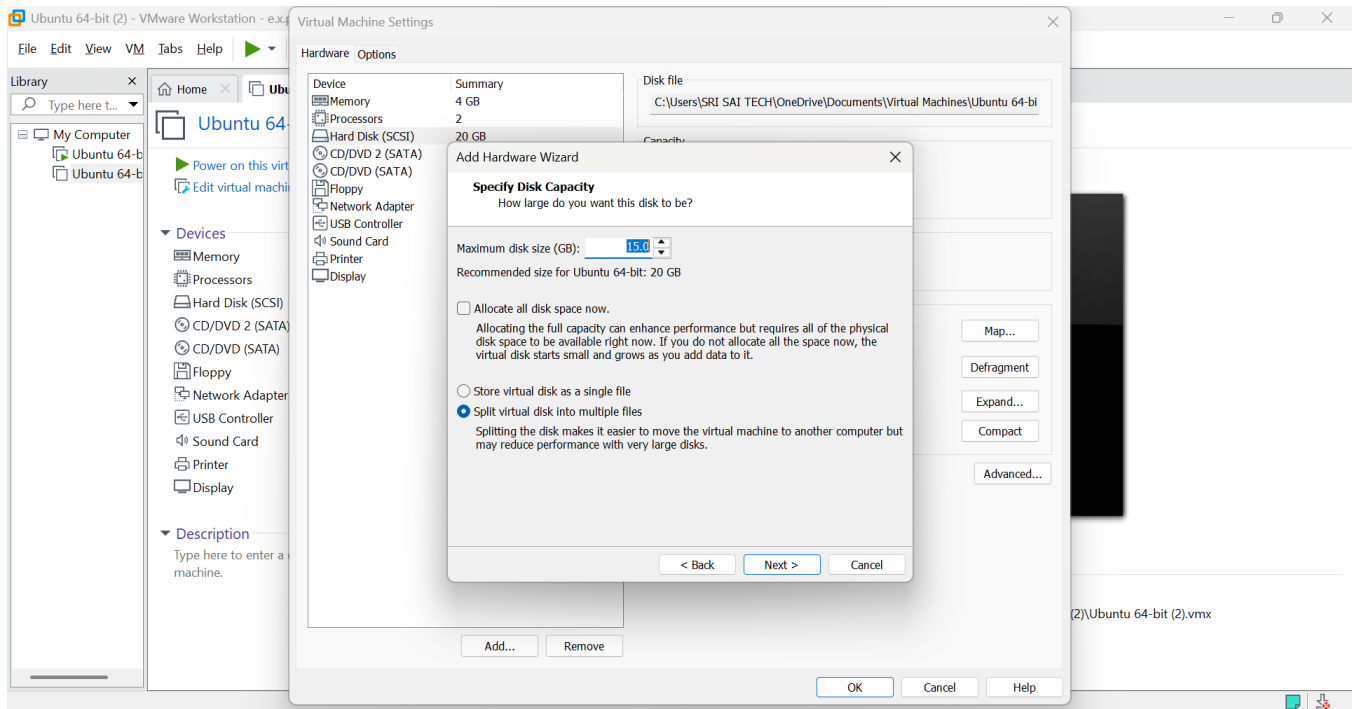


Step 4: Select Hard Disk → SCSI as the virtual disk type.



Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.





Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.

